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Millimeter Wave Radar Measurements: Distinguishing UAS and Birds Based on 60 GHz micro-Doppler Signatures

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***Abstract*—This work presents the results of measurements conducted on small drones and a bionic bird using a 60 GHz millimeter wave radar, analyzing their micro-Doppler characteristics in both time and frequency domains. In particular, we focus on their distinct nature of movement, i.e., rotating propellers and flapping wings, rather than relying on their materials. The time-series measurements show comparable differences in the phase of the samples as a result of micro-Doppler effects. Utilizing the collected measurement data, we develop neural network models to accurately differentiate between bionic birds and drones, having a significant potential for application in airports where precise object identification is essential. We adopt a convolutional neural network for detecting changes in the amplitude values and a convolutional long- and short-term memory for identifying the phase difference between the drone and bird signatures. The results reveal that distinguishing between small drones and birds can be done based on the phase difference of the scattered radar signals, even with a high noise variance.**

***Index Terms*—Classification, detection, drones, micro-Doppler, millimeter-wave, radar, UAV, UAS.**

I. Introduction

Due to their lower cost and increased accessibility, unmanned aerial systems (UAS), commonly known as drones, have found applications in diverse fields including agriculture, mapping, forensics, inspection, and cinematography, among others. The availability of commercially accessible drones enables various organizations to enhance task efficiency and workflow, leading to cost reductions. The drone industry has experienced significant growth in recent years, with emerging applications like drone deliveries and transportation gathering widespread attention. However, drones can be used for malicious purposes, posing a significant threat to security and privacy [1]. Their potential for carrying hazardous materials, conducting surveillance without authorization, and interfering with critical infrastructure, like airport areas, highlights the need for effective detection measures. The most widely spread commercial drones have a small form factor with a size smaller than 1 m, hence representing a bigger threat due to the challenge in detecting them [2].

A significant challenge in detecting small drones arises from the false alarm issue, as drones may have a similar size to birds. This is a particularly critical problem in airport zones, where drones pose a potential threat to aircraft safety during the take-off and landing process. The similar sizes and flight patterns of birds and drones can lead to fault detection, complicating the task of the deployed security systems. To avoid this problem, classifying drones and birds with high accuracy in various environments becomes a critical issue. Advanced sensing and analytical methods, such as machine learning algorithms capable of processing complex data from radar, acoustic sensors, and optical systems, become crucial. There have been studies in the literature to distinguish drones and birds using video analysis [3]–[5]. Commonly, proposed classification models are based on deep neural networks applied on video frames or images [6], [7]. In [8], the information on flight patterns for drones and birds is used for classification instead of video or image data.

Despite numerous efforts to classify drones and birds,

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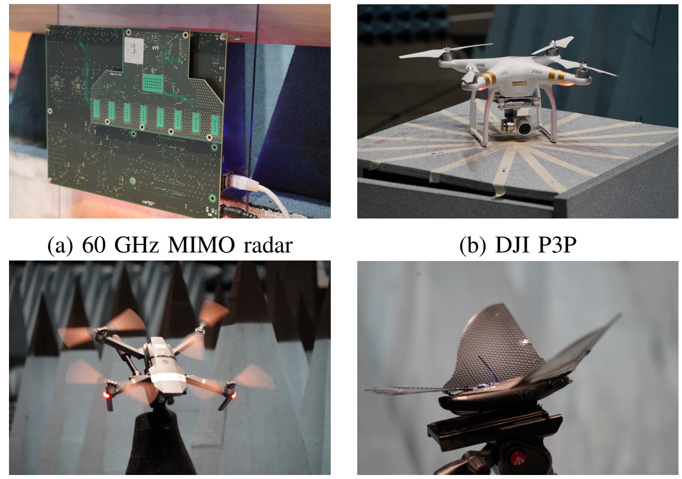
image-based classification causes false alarms owing to the failure to discern drones and birds. The challenges of image-based methods stem from several factors, including variable lighting conditions, obstructions in the line of sight, and the small size and high mobility of drones, which can closely mimic the flight patterns of birds. Radar detection emerges as a beneficial solution to these challenges for several reasons. Firstly, radar systems are less affected by visual impairments such as poor lighting or weather conditions, allowing for continuous operation, day and night, and in various weather scenarios. Second, radar can provide detailed information on the velocity, range, and angular position of an object, allowing a more nuanced analysis of its flight behavior compared to image-based systems. This information can be crucial in distinguishing between drones and birds based on their flight patterns, speeds, and altitude changes. Additionally, advanced radar technologies, such as millimeter-wave radars, offer higher resolution due to the large available bandwidth, enhancing the ability to detect small objects. Radar data is recognized for its substantial information regarding the micro-Doppler effect associated with drone movement [9]. Thus, in this work, we utilize micro-Doppler signatures extracted from bionic birds and drones as the basis for target recognition. We performed radar measurements and used the micro-Doppler radar to build machine machine-learning classification model for target discrimination. In the literature, several authors have investigated the micro-Doppler phenomenon [10]–[12] for separating small objects.

In particular, in [13] polarimetric signatures in the S-band are utilized to separate drone and bird by many polarimetric variables, and a Nearest-neighbor classifier is suggested. However, the prior work does not consider the micro-Doppler effect observed at different angles using millimeter wave radar. Furthermore, there has been little research to propose applications that employ neural networks for classification. This study presents an analysis of the measurements conducted using a millimeter wave MIMO radar from various angles. Based on the results, we observe distinct features of the data and discuss neural network models for target classification based on the micro-Doppler signatures. In addition, the robustness of the trained classification model is discussed for cases of leveraging echo amplitudes and phases separately. Therefore, we propose exploiting the phase signatures of drones and birds to build a more reliable classification model.

II. Measurement Campaign and Dataset

The millimeter wave radar utilized in this work has been developed at VTT Technology Research Centre of Finland Ltd, in Espoo, Finland, and is based on a 60 GHz frequency-division multiplexed frequency-modulated continuous-wave (FMCW) MIMO radar sensor [14]. The millimeter wave MIMO radar have a bandwidth of 6 GHz. The radar's architecture allows for significant scalability, with the ability to accommodate a large number of channels by incorporating multiple chips into the system. This scalability is facilitated by the design of separate TX and RX chips, which also helps reduce TX-RX leakage and offers more flexibility in antenna layout. A single external voltage-controlled oscillator (VCO) and phase-locked loop (PLL) distribute the local oscillator (LO) signal across all chips, ensuring good phase noise correlation. The radar system employs frequency-division multiplexing, transmitting non-overlapping frequencies from each transmitter, which enables the separation of signals from different transmitters at the receiver. This approach is advantageous for tracking moving targets and for applications requiring precise phase measurements. The detailed description of the radar initially developed can be found in [14]. In the current measurements, we utilize an improved version of the millimeter wave MIMO radar that has eight receivers and eight transmitters (Fig.1, a). Each transmitter transmits a frequency sweep that is modulated to a different frequency so that they can be separated from each other at the receivers.

In this study, measurements are done in an anechoic chamber located at VTT, Finland. The measurements are performed for 1 second with 1 kHz pulse repetition frequency. Two scenarios were considered for this measurement campaign: 1) no movement and 2) rotating propellers/flapping wings. The measurements are made for different azimuth angles, while the objects are located on a flat surface. The main idea was to obtain sufficient data that would replicate a more realistic scenario when the orientation of an object is not known. Next,



(a) 60 GHz MIMO radar (b) DJI P3P

(c) DJI Mavic (d) Bionic bird

Fig. 1: FMCW 60 GHz MIMO radar and objects under test

TABLE I: Measurement setup

Object location	**Object**	**Azimuth angles, [°]**	**Elevation angles, [°]**
Box, flat surface	DJI Mavic DJI P3P Bionic bird	0°, 22.5°, 45°, 67.5°, 90°, 112.5°, 135°, 167.5°, 180°	0°
Tripod allowing tilting	DJI Mavic DJI P3P Bionic bird	0°, 45°, 90°	-45°, 0°, 45°

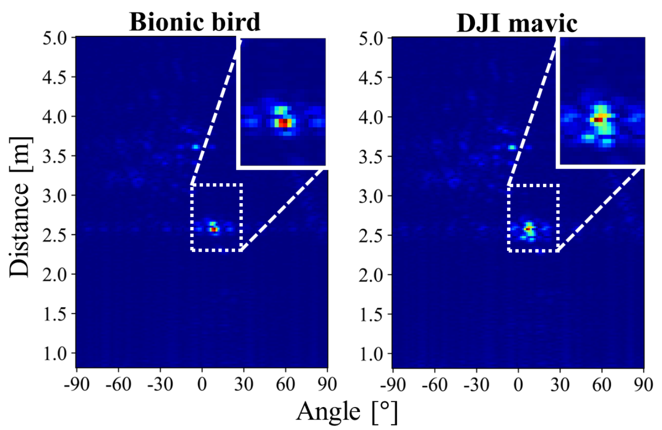
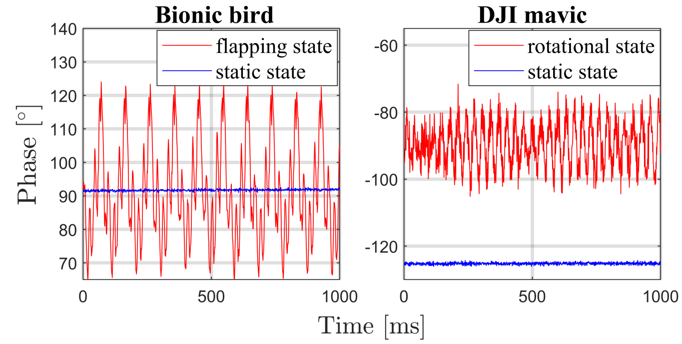
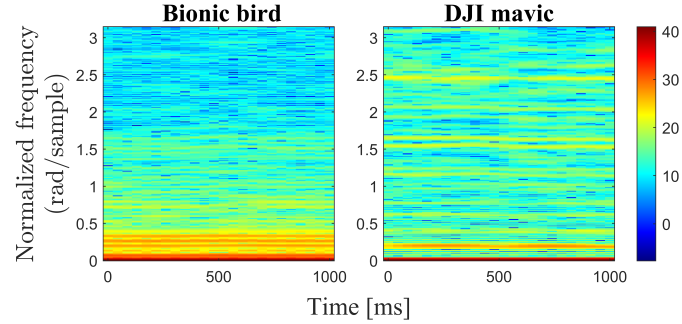


Fig. 2: Snapshot of the processed 2D radar data for bird and drone.



(a) Phase samples at locations of a propeller and wing



(b) Spectrograms

Fig. 3: Phase signatures over time and frequency

to simulate the view of the object from different elevation angles, the objects were placed on a tripod, allowing the drone and bird to tilt at various angles. The measurements were repeated for a combination of several azimuth and elevation angles when the propellers were rotating or stationary and when the wings were flapping or not. The measured angles for the test objects are summarized in Table I.

Several objects were considered during the measurement campaign. Specifically, we utilized drones such as the DJI Mavic and the DJI Phantom 3 Professional, chosen for their identical size to the available bionic bird model (Fig. 1, b-d). This selection was based on our research emphasis on different types of movement rather than material properties, aligning with our objective to explore motion characteristics for identification purposes. While the consideration of real birds in our measurements represents a logical next step, it introduces significant challenges, primarily due to the unpredictability and variability of their flight paths. Therefore, we focus here on the datasets obtained in the anechoic chamber and verify the possibility of the developed detection algorithms. By establishing a solid foundation of understanding based on controlled conditions, we can better assess the feasibility and adaptability of these algorithms in real-world scenarios.

We process the raw measurements to 2D radar data, which is formed by demodulating the MIMO channels, assembling the measurements into a linear virtual array, and taking the fast Fourier transform (FFT) to form the image [15]. From the result of FFT, we obtain 2D radar image data with the square of its amplitudes and 2D radar phase data by taking its angles. Fig. 2 shows the processed 2D radar images for both drone and bird, where the X and Y axes are the horizontal angle and distance from the objects. Since there are many redundant parts in the processed 2D radar data, as shown in Fig. 2, we only take data surrounding the detected drone or bird. Thus, the size of the final 2D radar data is much smaller than that of the original 2D data.

These measurement and processed data sets are publicly available for download [1]. It is necessary to mention that this radar is aimed at short-range detection and provides high-resolution measurement data. The distance during the measurements was set as $2.5\,\mathrm{m}$. In reality, the radar can operate and detect such objects up to $80\,\mathrm{m}$ away.

III. Data Analysis and Classification Model

In this section, we analyze the measured data and discuss classification models using measured radar data. First, we analyze the micro-Doppler features to classify drones and birds by focusing on the phase variations over time. Second, we build two different classification models by training neural networks, each of which uses the information of echo amplitudes and phases showing distinct features in the time and frequency domains. Lastly, the durability of models is discussed by testing them through different levels of signal-to-noise ratio (SNR).

A. Phase Signatures of Drone and Bird

The phase information of the drone and the bird shows prominent micro-Doppler characteristics because the rotor rotation rate for the drone and the wing beat cycle for flapping birds are clearly distinct. In the measurement, we observed different phase variations over time for drones and birds at the positions of propellers and wings. Thus, these distinct signatures of phases can be utilized for classification.

[1]The measurement dataset processed into a 2D data is available for download at https://dx.doi.org/10.21227/c8r5-6h79

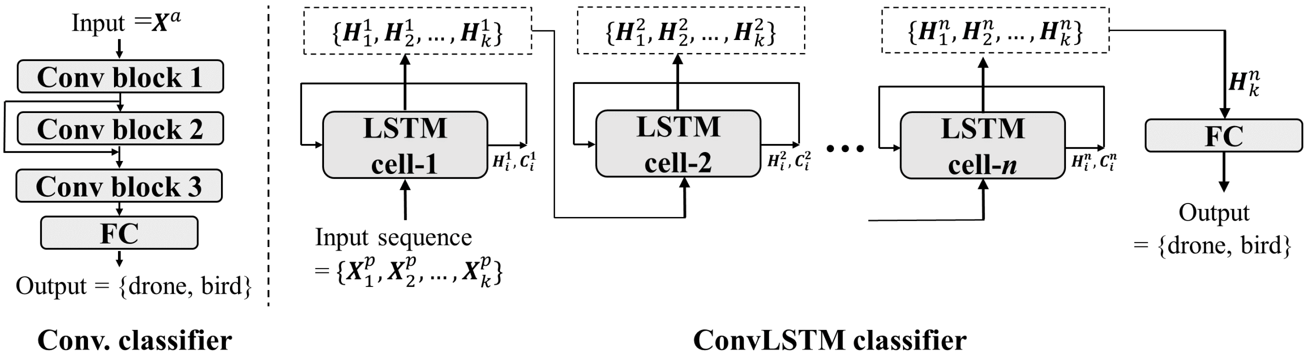


Fig. 4: Two classification models based on different structure of neural networks

Fig. 3a shows the variation of drone and bird phases at the location of the propeller and wings in the time domain. It can be easily observed that the phase variations increase when the drone and bird start to rotate and flap. The variations for drone and bird phases show that the peculiarities corresponding to the speed of the propeller and the bit rate of the wings are not the same.

To analyze the frequency content of the phases over time, we used *the spectrogram*, which is calculated by taking a short-term Fourier transform (STFT) for each block of a signal. The STFT is computed in the following way:

$$\text{STFT}\{x[n]\} = X(m,k) = \sum_{n=0}^{R-1} x[n+m]w[n]e^{-j\frac{2\pi}{N}kn}$$

where, $x[n]$ is the discrete phase value, $w[n]$ is the discrete window function, R is the length of the time window, N is the number of samples in the time window, and k and m are the indices of discrete frequency and position of the time window, respectively.

Fig. 3b shows drone and bird spectrograms in the form of a heat map, where each value indicates power spectral density (PSD) $(= |X(m,k)|^2)$. We observe that the high PSD values are concentrated at a lower sampling frequency for birds, while those for drones are placed at high frequencies. This result is correlated with the fact that the drone propeller cycle is shorter than that of the bird wings. Thus, through phase variations over time, we can capture the micro-Doppler characteristics of the drone propeller or the beat rate of the wings of the bird, and use them to discern the drone and the bird.

B. Classification Models

In this section, we discuss classification models that use echo amplitude and phase information from radar image data. First, we consider radar images where each pixel value is the echo amplitude and employ the image classification model using convolutional neural networks (CNN), which has one residual connection. In addition, since radar signals are measured at a distance close to the observer, to consider the attenuation effect at a distance far away, the echo amplitudes of the signals are multiplied by fraction factors randomly sampled from 0.1 to 1.0.

The picture on the left side of Fig. 4 describes the structure of our convolutional classifier (Conv. classifier), where there exist one residual connection between block 1 and 3. As shown, the input radar image $\boldsymbol{X}_i^a \in \mathbb{R}^{20\times 30}$ is fed into three convolutional blocks to capture the spatial correlation in the radar image, and the final fully connected neural network (FC) outputs the probabilities of drone and bird.

Next, we leverage the micro-Doppler characteristics using phase data. In the previous section, phase values over time for drones and birds show different peculiarities owing to the different micro-Doppler features occurring due to the speed in rotation of the propeller and beating rate of wings. A similar study was carried out with the [16] using 1D phase data measured from the $94\,\text{GHz}$ radar. However, this study has pronounced shortcomings in that the location of the propeller and wings must be estimated to capture micro-Doppler signatures in the 1D phase data. In other words, the detected images can be multiple pixels with a larger bandwidth within a certain range of distances. Additionally, due to the dynamic movement of drones and birds, finding a specific location of the propeller or wings is very challenging, which causes estimation errors. Therefore, rather than using 1D phase information, we propose exploiting 2D phase frame sequences measured in the spatial region of the radar image over some consecutive time interval. These 2D phase frame sequences are the same size as the radar images and include phase variations at the location of the propeller or wings, as well as at other locations.

To classify drone and bird based on the measured 2D phase frames, we use the Convolutional Long-Short-Term Memory (ConvLSTM) model, which has shown good performance in predicting future values of a time series of image frames [17] by capturing the spatiotemporal characteristics between time series images. Specifically, in this model, the convolutional layers capture the correlation in the spatial domain, while the LSTM layers learn the correlation between sequences of phases over a certain time period. More detailed key equations are given in [17].

The right side of Fig. 4 describes the whole structure of our ConvLSTM classifier. As shown in Fig. 4, the phase sequences over k consecutive time slots, $\{\boldsymbol{X}_i^p, \boldsymbol{X}_{i+1}^p, \cdots, \boldsymbol{X}_k^p\}$,

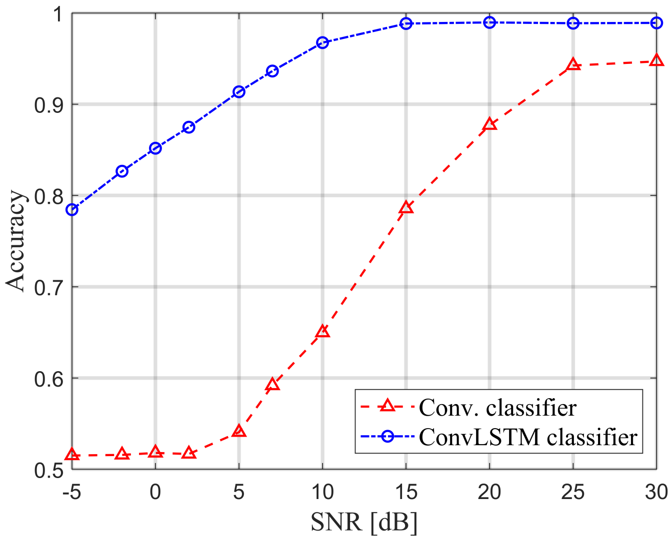


Fig. 5: Accuracy comparison for different SNR values

where $\mathbf{X}_i^p \in \mathbb{R}^{20\times 30}$, are fed into the LSTM cell, and then convolutional operations are applied inside the LSTM cell. When using multiple LSTM cells, the hidden states $\{\boldsymbol{H}_1, \boldsymbol{H}_2, \cdots, \boldsymbol{H}_k\}$, where $\boldsymbol{H}_i \in \mathbb{R}^{20\times 30}$, become the input tensors of the next LSTM cell. In the last LSTM cell, fully connected network output the final probabilities of drone and bird with the last hidden state $\boldsymbol{H}_k^n$.

The total number of parameters in Conv. classifier is 753.2K, while there are 768.1K parameters in ConvLSTM. All the detailed implementations of Conv. classifier and ConvLSTM are uploaded in [18].

C. Robustness of Classification

In this section, we evaluate the robustness of the proposed models when radar signals are distorted with different levels of signal-to-noise ratio (SNR). We assume that the signals propagated from a considerable distance have low SNR, and thus, in such a case, discerning drones and birds can be challenging. To simulate this, we add the zero-mean Gaussian noise $N(0, \sigma^2)$ to the amplitude values of the radar data, which leads to distortion of the phase values. As an evaluation metric, for 2D radar data $\boldsymbol{D} \in \mathbb{C}^{M\times N}$ and noise variance σ^2, we define SNR as follows.

$$\mathrm{SNR} = \frac{\frac{1}{MN}\sum_{i}^{M}\sum_{j}^{N}|\boldsymbol{D}_{i,j}|^2}{\sigma^2} \tag{1}$$

where $\boldsymbol{D}_{i,j} \in \mathbb{C}$ is the element of the 2D radar data $\boldsymbol{D}$.

Fig. 5 shows the classification accuracy of each model for different values of SNR. As the SNR values decrease, the performance of Conv. classifier deteriorates rapidly. In contrast, the classification accuracy of the ConvLSTM model is still greater than 85% even with $0\,\mathrm{dB}$ SNR. The performance gap on testing models with the distorted data is due to the fact that ConvLSTM captures the correlation in the spatial and time domains simultaneously, while the Conv. classifier only learns the spatial correlation. As we discussed in Sect. III-A, time-domain correlations of phase data are strongly correlated with the micro-Doppler properties of the drone and bird. Thus, we argue that exploiting different micro-Doppler characteristics is instrumental in building a noise-tolerant model to distinguish drone and bird even though the received signals are severally distorted.

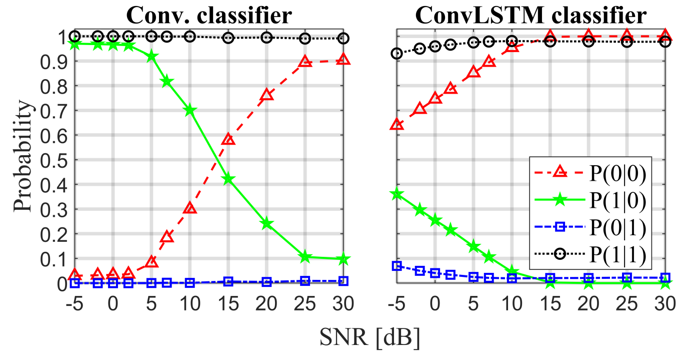


Fig. 6: Prediction probability of each label $\{0, 1\}$ given the true label $\{0, 1\}$ where 0 and 1 indicate bird and drone.

Next we analyze the conditional probability of the neural network models. Let a binary random variable $X = \{0, 1\}$ denote the true label and another binary random variable $\hat{X} = \{0, 1\}$ denote the predicted label. We use 0 for the label of the mechanical bird and 1 for label of the drone. We calculate the prediction probabilities given their true label (i.e., $P(\hat{X}|X)$). In Fig. 6, we show the probability of classification (i.e., P$(0|0)$, P$(1|1)$) and the probability of misclassification (i.e., P$(0|1)$, P$(1|0)$) over SNR.

We confirm that both models do not detect the true bird label at low SNRs with high probability. In other words, P$(0|0)$ reduces and P$(1|0)$ increases as the SNR decreases, while the true drone label is well detected for both models (P$(1|1)=1$ and P$(0|1)=0$). In particular, when the SNR is around $15\,\mathrm{dB}$, Conv. classifier fails to detect birds. The results in Fig. 6 imply that the Conv. classifier, which captures only spatial correlation, is very susceptible to noise and does not detect birds even with a high SNR value. Thus, we propose that ConvLSTM, which learns the micro-Doppler characteristics using phase information, should be employed as a noise-resilient classification model that does not require the specific location information of drone propellers or bird wings.

IV. Conclusion

In this work, we present the experimental results from tests conducted on two drone models and a bionic bird using the $60\,\mathrm{GHz}$ MIMO FMCW radar. The measurements are carried out for static and dynamic cases. In the dynamic measurements, specific attention was paid to capturing the rotation of drone propellers and the flapping motion of the bird's wings with the radar. The focus of this research is to distinguish between drones and birds effectively, thereby minimizing false detection rates. The analysis process involves classifying drones and birds by examining the differences in their phase measurements. To achieve this, we implemented a Convolutional Neural Network (CNN) for analyzing amplitude data and a Convolutional Long Short-Term Memory (Conv-LSTM) model for phase data analysis. Our findings highlight the critical role of phase information in the accurate identification of objects, particularly in environments with significant noise levels. The CNN model, in comparison, showed limitations in recognizing objects based solely on Radar Cross Section (RCS) measurements. The processed radar data is available in [19].

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